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Road network circuity in metropolitan areas

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Abstract. Circuity, the ratio of network to Euclidean distances, describes the directness of trips and the efficiency of transportation networks. This paper measures the circuity of the fifty-one most populated Metropolitan Statistical Areas (MSAs) in the United States and identifies trends in those circuities between 1990 and 2010. Overall circuity has increased between 1990 and 2010: random points have not only become farther apart in distance, their shortest network path has become more circuitous, suggesting that the more recently constructed parts of street networks are laid out more circuitously than older parts of the network. Over this period thirty-five MSAs experienced a statistically significant increase in circuity (six experienced a significant decrease). As expected, short trips are more circuitous than long trips. A new circuity distance-decay function describes how circuity varies with distance within metropolitan areas. The parameters of this function have changed from 1990 to 2010.

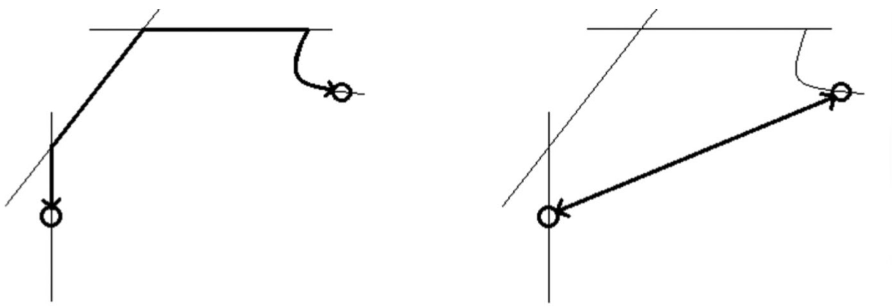
Keywords: circuity, directness, network structure, cities

Introduction

Road networks exploit economies of scale to increase travel speed at the expense of increased distance. The impact of topology on networks is well documented (Jiang, 2007; Jiang and Claramunt, 2004, Jiang et al, 2013). Bodies of water, mountains, hills, or even a lack of demand can increase deviations from the straight line. But, even without topological interference, road networks require greater distances than 'as the crow flies'. Circuity, which quantifies this deviation from a distance-minimizing straight line, is the ratio of network to Euclidean distance (Barthélemy, 2011; Barthélemy and Flammini, 2008; Derrible, 2012; Fisher and Hochmair, 2005; Levine et al, 2012; Levinson and El-Geneidy, 2009; Parthasarathi et al, 2010; 2012; Roth et al, 2010; Strano et al, 2012). Figure 1 shows the two distances. Circuity has been used to aid in the placement of centralized facilities (Forkenbrock and Foster, 1996; Goldman, 1972), selecting highway alignments (Jha, 2003), and organization of shipping logistics (Popken, 2006; Southworth and Peterson, 2000; Uester and Kewcharoenwong, 2011). Even in the shipping industry, ocean currents and inclement weather complicate the ideal circuity for any trip. Through these studies and others, it is clear that implications of network circuity abound. This research focuses on road networks, which serve the vast majority of trips in the US and elsewhere.

On road networks, previous research has approximated circuity values to average around 1.2 (Newell, 1980). Circuity has been shown to vary by country (Ballou et al, 2002), likely due to variance in network density. These properties and more affect networks and their connectivity.

A perfectly connected network yields circuity values of 1.0 for all possible trips. The cost for such a network would be immense. Trade-offs exist between construction, operations, and maintenance costs, and the utility provided by the network (Carmi et al, 2008; Werner, 1968). The problem of circuity is further complicated by flows exceeding traffic capacities



(a) (b)

Figure 1. A comparison of (a) network and (b) Euclidean routes, both used in the calculation of circuity.

on links, leading to congestion and further deviation in search of other routes that offer travel time savings.

This study measures the circuity for road networks in the fifty-one most populous Metropolitan Statistical Areas (MSAs) in the United States for 1990, 2000, and 2010. If circuity changes, this implies that the design of networks has changed or the use of networks has changed, or both. If circuity increases, it implies networks are less efficient from a shortest distance path perspective (though they may be as or more efficient from a shortest travel time path perspective, as not all links have the same speeds). This study also examines how circuity varies with the length of origin–destination pairs (OD pairs), estimating a measure of distance decay for circuity.

Data

Census data

MSAs as defined in 2009 by the census for 2010 were used as a basis for this study, ensuring a consistent geography for all three points in time. MSA boundaries change decennially. We considered using the definitions from 1990 and 2000 with the census road network data from 1990 and 2000, respectively, but decided against for two reasons. First, the definitions change drastically with some MSAs. For instance, in 1990 the Phoenix MSA definition included only Maricopa County. By 2009 the MSA definition also included Pinal County. Another issue concerned the definition of Consolidated Metropolitan Statistical Areas (CMSAs) as opposed to MSAs. This can be contrasted with the census nomenclature used in 1990, where within MSAs there were Primary Metropolitan Statistical Areas (PMSAs), markedly different from the definitions in 2009. A good example of this is Pittsburgh where in 1990 the area was divided into two PMSAs and in 2009 the counties were listed together in one MSA. In 2009 the census did not define CMSAs nor PMSAs, but instead defined Metropolitan Divisions within MSAs.

Sampling

Sampling of origins and destinations used in circuity calculations occurs randomly across the network. Points could fall on any polygon representing a link. It is worth noting that GIS files generated by the census can have errors, and it is possible that low-volume exurban roads are less well represented in older GIS files than in new ones. In calculations the distance from the network link nearest the point was used. For areas with no roads, no points would be generated. Figure 2 depicts a portion of the Miami MSA with sample origins. As can be seen, points fall on the polygon containing the center lines of roads, such as in rural areas, so the distribution of points is proportional to the location of roads.

The 2000 and 2010 MSA networks were projected onto state planar coordinates, North American Datum (NAD) 1983. Files for the 1990 road networks were created

Figure 2. [In color online.] A selected portion of the Miami road network with generated origins overlaid in red.

using NAD 1927, and a conversion to 1983 was required. Using GIS software (ArcGIS), 500 points generated over the MSA represented randomly located home origins, and 100 points generated represented work destinations. The product of the home and work points, about 50 000 OD pairs, representing 'trips', were generated for each MSA. Ideally, 50 000 OD pairs, would be generated, though it was common that a few points would fail to have network paths connecting them, resulting in a total generation of trips slightly under 50 000 for each year for each MSA. The shortest distance path on the road network of the MSA was calculated for each OD pair. Euclidean distances were also computed.

The reproducibility of results is essential. The OD matrix of 500 home locations and 100 work locations is the result of careful sensitivity analysis. To ensure reproducibility, many simulations were run on the road network of Miami, Florida. Miami was chosen because it is a large MSA that has relatively few (three) counties (Broward, Miami-Dade, and Palm Beach) making it relatively simple to manage. Five simulations were run for each OD matrix of 2000, 8000, 50 000, and 200 000 trips, twenty simulations in total. Unweighted circuity values were used as a measure, as these are the raw data. The results of these iterative tests can be seen in table 1. Obviously from each set of generated OD pairs, the unweighted circuity

Table 1. Sampling iterations (trips) performed on Miami, Florida Metropolitan Statistical Area (MSA).

Trips	Matrix	C	σ
2 000	[100 × 20]	1.198185	0.012191
8 000	[200 × 40]	1.180127	0.010263
50 000	[500 × 100]	1.184671	0.006425
200 000	[1000 × 200]	1.186153	0.009032

value will differ slightly from test to test. After a few trials with each matrix the results of each matrix size are compared.

The standard deviations (σ) of these results were used to determine what size matrix to use for testing the remaining MSAs. Since the standard deviations of unweighted circuity results as calculated in equation (2) below do not decrease beyond 50 000 trips generated, the sample size chosen for all MSAs was 50 000 trips. With a standard deviation (σ) of 0.0064, this creates a 95% confidence interval of ±0.0126 in the unweighted circuity value. Each of the fifty-one MSAs has matrices generated for 1990, 2000, and 2010. That is 153 iterations, and with a 95% confidence interval it is expected that approximately seven or eight of those iterations would fall outside the aforementioned confidence interval. From this testing, it was determined that larger sample sizes were unnecessary.

Weighting factors

To ensure the measures are reasonable, weighted circuity weights the network OD pairs by the likelihood of that distance appearing in actual home–work commutes.

Such data are collected in the Federal Highway Administration, US Department of Transportation National Household Travel Survey (NHTS). The NHTS was conducted most recently in 1995, 2001, and 2009. Data from 1995 were used for 1990 trip length distribution because the way data were collected in 1990 made it incomparable with data from subsequent years. Those surveys are overlaid with the data generated from the road networks of 1990, 2000, and 2010, respectively.

The geographic area for weighting needs to be considered because some MSAs are smaller than others and cannot have trips which reach a given distance and remain within the MSA. Those trips likely originated from (or were destined to) locations outside the MSA. Any trips simulated in excess of 60 km were excluded. An interval size (s) of 5 km was chosen based on ensuring a sufficient sample size in each MSA. A threshold (S) of 60 km included more than 95% of work trips in the tested MSAs. For some MSAs, data were not collected by the NHTS for all three surveys, and proxies (values from the nearest year for which data were collected) were used. MSAs that did not have survey data for the 1995 survey are: Atlanta, Grand Rapids, Honolulu, Jacksonville, Louisville, Providence, and Raleigh. For those MSAs, survey data from 2001 were used as weights. The only MSA that did not have survey data for just the 2009 survey was Honolulu, for which survey data from 2001 were used. Adjoining MSAs were combined in the NHTS in some years: specifically, Baltimore and Washington, DC were listed separately in 1995, but were combined for the 2001 and 2009 surveys.

Circuity definitions

The circuity of an individual trip is calculated according to:

$$C = \frac{D_{\mathrm{N}}}{D_{\mathrm{E}}} \, , \tag{1}$$

where D_{N} is minimum traversable distance on a network, and D_{E} is the Euclidean distance (straight-line between origin and destination). Equation (1) represents the calculation of circuity of an individual trip. In subsequent equations, this calculation will be performed over many trips and weighted according to actual home–work trips in the most populated US metropolitan areas. This method is consistent with previous literature (Levinson and El-Geneidy, 2009; Popken, 2006).

Unweighted circuity of an MSA m is calculated as:

$$C_{u,m} = \frac{\sum_{m} D_{\mathrm{N}}}{\sum_{m} D_{\mathrm{E}}} \, , \tag{2}$$

where
$C_{u,m}$ is the average unweighted circuity in MSA m,
$\sum_m D_{\mathrm{N}}$ is the sum of the network distances between all OD pairs in the sample,
$\sum_m D_{\mathrm{E}}$ is the sum of the Euclidean distances between all OD pairs in the sample.

While this appears to be indicative of the circuity of a road network, it does little to show the circuities of actual commutes in MSAs. To show this, weighting must occur.

Equation (2) does not calculate the average circuity experienced by commuters within MSAs; longer trips are heavily favored in calculation. So we can compute the circuity in intervals:

$$C_{u,i} = \frac{\sum_i D_{\mathrm{N},i}}{\sum_i D_{\mathrm{E},i}} , \tag{3}$$

where
$C_{u,i}$ is the average unweighted circuity in interval i,
$\sum_i D_{\mathrm{N},i}$ is the sum of the network distances between all OD pairs in interval i,
$\sum_i D_{\mathrm{E},i}$ is the sum of the Euclidean distances between all OD pairs in interval i,

where we sum over a 5 km interval i instead of the entire MSA. Using those intervals, we can apply weights and compute a weighted circuity for the region:

$$C_{w,m} = \frac{\sum_{i=1}^{I} T_i C_{u,i}}{\sum_{i=1}^{I} T_i} , \tag{4}$$

where
$C_{w,m}$ is the weighted circuity of trips less than or equal to threshold S, for MSA m,
T_i is the number of home–work trips in interval i,
$C_{u,i}$ is the unweighted circuity of trips in interval i,

and interval size $s = 5$ km, threshold $S = 60$ km, and the total number of intervals $I = S/s = 12$.

Results

Table 2 shows the unweighted circuities as calculated in equation (2) for trips up to 60 km in fifty-one MSAs for 1990, 2000, and 2010. Student t-tests were performed between data from 1990 and 2010. Respective p-values and confidence intervals are tabulated in the last two columns.

Similarly, table 3 shows the results of commute trip length frequency weighted circuities from equation (4). Calculated in the last row of table (3) is the average (weighting all fifty-one cities equally) for each year analyzed (1990, 2000, 2010). In these averages, a clear increasing trend can be seen.

Table 4 summarizes the trends in circuity. For example, 1990–2000–2010 means that the weighted circuity was lowest in 1990, higher in 2000, and highest in 2010. As can be seen from table 4, the first three possibilities listed have 1990 with a lower weighted circuity value than 2010. Of the fifty-one MSAs, forty-one yielded statistically significant results, thirty-five showed statistically significant increases, and six showed statistically significant decreases. For statistical significance values, see the last two columns in tables 2 and 3.

The weighted circuity value in table 3 is almost always lower than its unweighted counterpart in table 2. Exceptions were Honolulu and Salt Lake City for all three analyzed

Table 2. Unweighted circuities (1990–2010).

Metropolitan Statistical Area	1990	2000	2010	t-statistic	p-value	Confidence interval (%)
Atlanta	1.213	1.213	1.217	5.541	3.02×10^{-8}	99
Austin	1.294	1.289	1.293	2.089	3.67×10^{-2}	95
Baltimore	1.215	1.217	1.232	4.985	6.21×10^{-7}	99
Boston	1.192	1.192	1.192	−3.630	2.84×10^{-4}	99
Buffalo	1.172	1.186	1.179	7.809	5.80×10^{-15}	99
Charlotte	1.218	1.227	1.220	0.365	7.15×10^{-1}	0
Chicago	1.189	1.187	1.191	1.653	9.83×10^{-2}	90
Cincinnati	1.280	1.271	1.272	2.685	7.24×10^{-3}	99
Cleveland	1.164	1.166	1.166	−3.538	4.03×10^{-4}	99
Columbus	1.190	1.191	1.190	0.266	7.90×10^{-1}	0
Dallas-Fort Worth	1.226	1.227	1.236	3.521	4.31×10^{-4}	99
Denver	1.326	1.320	1.342	3.131	1.74×10^{-3}	99
Detroit	1.189	1.198	1.192	4.507	6.59×10^{-6}	99
Grand Rapids	1.233	1.239	1.240	−0.537	5.91×10^{-1}	0
Hampton Roads	1.338	1.306	1.307	5.001	5.72×10^{-7}	99
Hartford	1.214	1.227	1.228	3.856	1.15×10^{-4}	99
Honolulu	1.467	1.452	1.443	3.425	6.15×10^{-4}	99
Houston	1.262	1.261	1.263	2.396	1.66×10^{-2}	95
Indianapolis	1.185	1.203	1.198	4.713	2.45×10^{-6}	99
Inland Empire	1.315	1.335	1.351	5.574	2.50×10^{-8}	99
Jacksonville	1.309	1.310	1.305	4.710	2.48×10^{-6}	99
Kansas City	1.252	1.254	1.247	7.475	7.79×10^{-14}	99
Las Vegas	1.327	1.338	1.316	0.333	7.39×10^{-1}	0
Los Angeles	1.242	1.230	1.236	5.700	1.20×10^{-8}	99
Louisville	1.295	1.296	1.307	−6.401	1.55×10^{-10}	99
Memphis	1.257	1.272	1.254	2.233	2.55×10^{-2}	95
Milwaukee	1.174	1.178	1.168	1.173	2.41×10^{-1}	0
Minneapolis-St.Paul	1.230	1.223	1.229	2.090	3.67×10^{-2}	95
Nashville	1.287	1.326	1.285	0.181	8.56×10^{-1}	0
New Orleans	1.276	1.301	1.335	3.231	1.23×10^{-3}	99
New York	1.233	1.219	1.212	1.094	2.74×10^{-1}	0
Oklahoma City	1.263	1.253	1.247	3.356	7.90×10^{-4}	99
Orlando	1.306	1.270	1.320	3.853	1.17×10^{-4}	99
Philadelphia	1.169	1.176	1.171	3.479	5.04×10^{-4}	99
Phoenix	1.262	1.301	1.278	2.082	3.74×10^{-2}	95
Pittsburgh	1.255	1.251	1.270	−1.158	2.47×10^{-1}	0
Portland	1.405	1.418	1.441	2.748	6.00×10^{-3}	99
Providence	1.226	1.226	1.234	13.346	1.35×10^{-40}	99
Raleigh	1.199	1.208	1.208	8.596	8.37×10^{-18}	99
Rochester, NY	1.204	1.194	1.209	1.453	1.46×10^{-1}	0
Sacremento	1.435	1.456	1.396	−7.771	7.88×10^{-15}	99
Salt Lake City	1.355	1.436	1.382	3.556	3.77×10^{-4}	99
San Antonio	1.312	1.298	1.294	−2.237	2.53×10^{-2}	95
San Diego	1.413	1.386	1.393	4.161	3.17×10^{-5}	99
San Francisco	1.373	1.356	1.393	7.955	1.82×10^{-15}	99
San Jose	1.395	1.421	1.403	0.092	9.27×10^{-1}	0
Seattle	1.408	1.386	1.409	4.248	2.16×10^{-5}	99
South Florida	1.218	1.231	1.238	6.313	2.76×10^{-10}	99
St. Louis	1.317	1.312	1.323	1.065	2.87×10^{-1}	0
Tampa	1.252	1.242	1.269	10.425	1.97×10^{-25}	99
Washington	1.289	1.264	1.275	−2.514	1.19×10^{-2}	95
Average	1.271	1.273	1.274			

Table 3. Weighted circuities (1990–2010).

Metropolitan Statistical Area	1990	2000	2010	t-statistic	p-value	Confidence interval (%)
Atlanta	1.287	1.304	1.318	8.388	5.02×10^{-17}	99
Austin	1.371	1.359	1.382	2.401	1.63×10^{-2}	95
Baltimore	1.296	1.288	1.316	7.273	3.55×10^{-13}	99
Boston	1.278	1.276	1.263	–5.123	3.01×10^{-7}	99
Buffalo	1.237	1.256	1.258	12.311	8.38×10^{-35}	99
Charlotte	1.292	1.300	1.294	0.522	6.01×10^{-1}	0
Chicago	1.266	1.257	1.275	2.567	1.03×10^{-2}	95
Cincinnati	1.347	1.352	1.362	3.817	1.35×10^{-4}	99
Cleveland	1.253	1.239	1.243	–5.398	6.75×10^{-8}	99
Columbus	1.280	1.280	1.281	0.366	7.14×10^{-1}	0
Dallas-Fort Worth	1.297	1.298	1.319	4.462	8.14×10^{-6}	99
Denver	1.361	1.343	1.374	3.360	7.80×10^{-4}	99
Detroit	1.256	1.263	1.264	6.510	7.53×10^{-11}	99
Grand Rapids	1.299	1.294	1.297	–0.769	4.42×10^{-1}	0
Hampton Roads	1.352	1.370	1.368	7.287	3.20×10^{-13}	99
Hartford	1.282	1.290	1.288	5.484	4.18×10^{-8}	99
Honolulu	1.420	1.407	1.435	4.176	2.96×10^{-5}	99
Houston	1.338	1.329	1.347	2.674	7.50×10^{-3}	99
Indianapolis	1.277	1.292	1.284	6.517	7.21×10^{-11}	99
Inland Empire	1.323	1.347	1.368	5.957	2.58×10^{-9}	99
Jacksonville	1.347	1.380	1.362	5.913	3.37×10^{-9}	99
Kansas City	1.294	1.310	1.319	9.969	2.13×10^{-23}	99
Las Vegas	1.316	1.329	1.319	0.399	6.90×10^{-1}	0
Los Angeles	1.274	1.276	1.307	7.893	2.98×10^{-15}	99
Louisville	1.397	1.370	1.358	–8.518	1.64×10^{-17}	99
Memphis	1.319	1.340	1.331	3.309	9.37×10^{-4}	99
Milwaukee	1.235	1.254	1.238	1.822	6.84×10^{-2}	90
Minneapolis-St.Paul	1.301	1.290	1.310	2.842	4.49×10^{-3}	99
Nashville	1.365	1.414	1.366	0.220	8.26×10^{-1}	0
New Orleans	1.365	1.363	1.390	4.531	5.88×10^{-6}	99
New York	1.293	1.298	1.300	1.339	1.81×10^{-1}	0
Oklahoma City	1.297	1.315	1.316	4.846	1.26×10^{-6}	99
Orlando	1.358	1.329	1.380	5.047	4.49×10^{-7}	99
Philadelphia	1.254	1.271	1.269	4.717	2.39×10^{-6}	99
Phoenix	1.310	1.338	1.331	2.220	2.64×10^{-2}	95
Pittsburgh	1.380	1.377	1.374	–1.419	1.56×10^{-1}	0
Portland	1.454	1.462	1.484	3.015	2.57×10^{-3}	99
Providence	1.296	1.296	1.322	15.283	1.14×10^{-52}	99
Raleigh	1.267	1.287	1.296	14.685	9.07×10^{-49}	99
Rochester, NY	1.277	1.267	1.281	2.202	2.77×10^{-2}	95
Sacremento	1.449	1.464	1.4111	–8.553	1.21×10^{-17}	99
Salt Lake City	1.341	1.414	1.378	4.773	1.82×10^{-6}	99
San Antonio	1.410	1.384	1.394	–2.615	8.94×10^{-3}	99
San Diego	1.411	1.399	1.437	4.614	3.95×10^{-6}	99
San Francisco	1.385	1.402	1.439	9.200	3.63×10^{-20}	99
San Jose	1.410	1.430	1.411	0.110	9.12×10^{-1}	0
Seattle	1.425	1.398	1.465	5.132	2.87×10^{-7}	99
South Florida	1.289	1.299	1.307	7.321	2.48×10^{-13}	99
St. Louis	1.380	1.369	1.389	1.334	1.82×10^{-1}	0
Tampa Bay	1.313	1.317	1.334	12.623	1.68×10^{-36}	99
Washington	1.371	1.349	1.354	–3.360	7.79×10^{-3}	99
Average	1.327	1.332	1.339			

Table 4. Frequencies of statistically significant circuity trends.

Year (ascending)	Count
1990–2000–2010	14
1990–2010–2000	8
2000–1990–2010	13
2000–2010–1990	3
2010–1990–2000	1
2010–2000–1990	2
Total Metropolitan Statistical Areas with significant changes	41
Total Metropolitan Statistical Areas	51

years, the 1990 versions of San Diego and Las Vegas, and the 2000 road network of Las Vegas. This tendency is to be expected, as weighting the circuity values tends to have shorter trips than unweighted circuity.

The effect of trip length on circuity

In figure 3 trends for trips generated within the Minneapolis MSA are shown. C_i shows an unweighted circuity value according to equation (3) over a certain network distance interval i for each decade. We can similarly draw this for network travel time intervals (C_t). Circuity decreases as the travel distance and time increase. These observations corroborate previous findings that higher circuities are experienced with shorter commutes (Levinson and El-Geneidy, 2009).

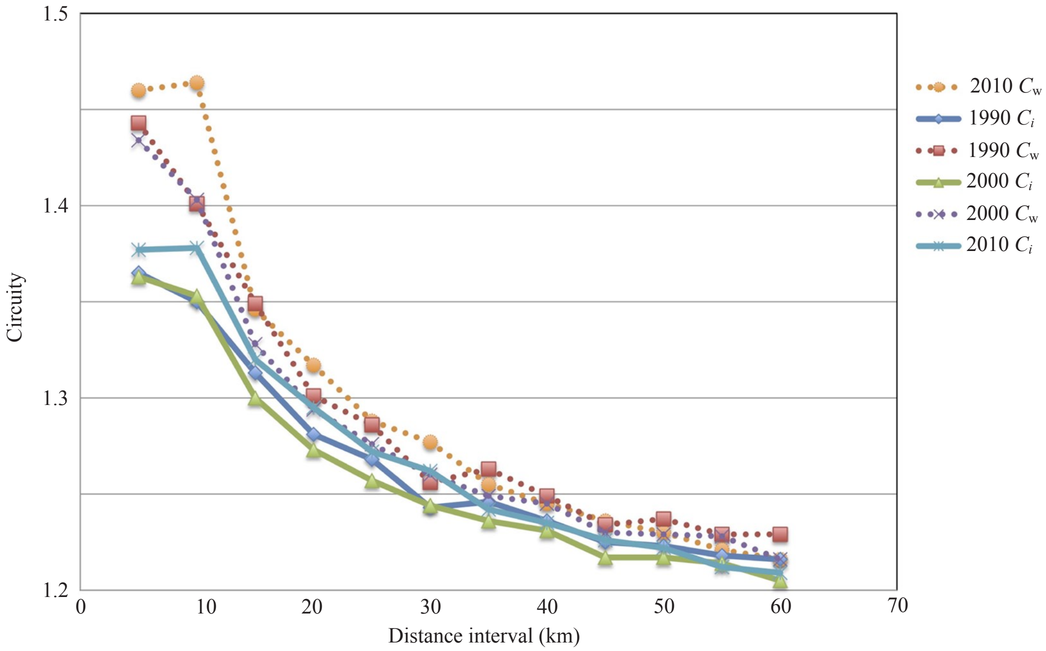


Figure 3. [In color online.] Circuity (C_w—weighted, C_i—unweighted) by distance interval in Minneapolis.

This curve can be modeled exponentially (Levinson, 2012). In searching for the best exponential fit, when looking at unweighted circuity and travel time, linear fits were explored for all fifty-one MSAs in the study on linear plots, double-log plots, and both semilog plots.

The double-log plot gave the best fit (r^2) for all three years analyzed in the study (1990, 2000, and 2010). The corresponding equation follows:

$$C_t = e^{\beta} t^{\zeta} , \qquad (5)$$

where
C_t is the unweighted circuity for some timeband (t),
ζ is the circuity decay parameter to be estimated,
β is the constant to be estimated.

Table 5 summarizes the results. As can be seen, ζ decreases on average from −0.035 to −0.039 from 1990 to 2010, indicating that if the βs were constant, circuity would be lower in 2010 than 2000, which would be lower than 1990. However, β also varies slightly over time, so the estimated average circuity is lowest in 1990 for trips under 60 minutes, followed by 2010, with 2000 having the highest values, as shown in figure 4.

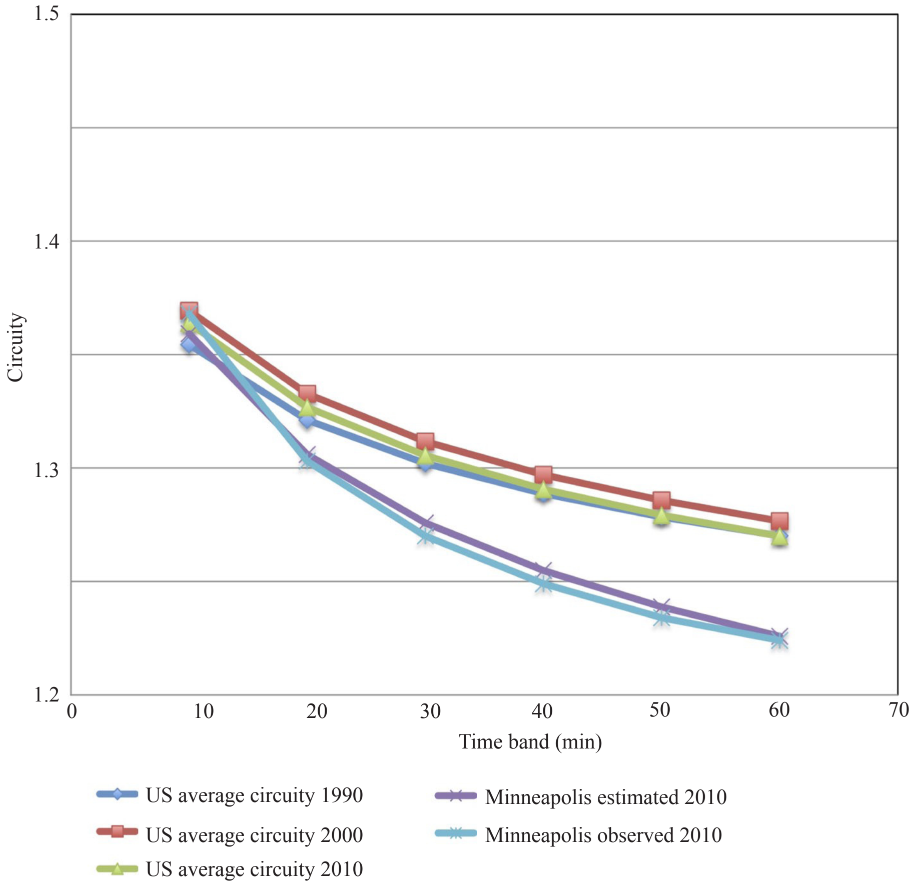


Figure 4. [In color online.] Circuity (C_t) by time interval, US average and Minneapolis. An average weighted circuity function averaged over all 51 MSAs studied in 1990, 2000, and 2010.

Figure 5 shows the ability of equation (5) to model circuity data. The plotted points in figure 5 show US averages for circuity in each time interval. The plotted curve is a representation of the circuity using equation (5). Error bars are plotted to show the distribution of data points from all fifty-one MSAs. As can be seen, there is high overlap of the circuities between cities, and the difference between cities is much greater than the difference between years.

Table 5. Summary table of ζ , β, and respective r^2 values.

Metropolitan Statistical Area	1990			2000			2010		
	ζ	β	r^2	ζ	β	r^2	ζ	β	r^2
Atlanta	−0.05823	0.4321	0.997	−0.06512	0.4687	0.996	−0.06962	0.4777	0.991
Austin	−0.06030	0.5047	0.976	−0.04778	0.4577	0.951	−0.06266	0.5037	0.988
Baltimore	−0.05200	0.4102	0.984	−0.05241	0.4167	0.971	−0.05439	0.4344	0.915
Boston	−0.05162	0.3879	0.953	−0.04943	0.3810	0.948	−0.05133	0.3865	0.961
Buffalo	−0.03442	0.2957	0.979	−0.03864	0.3299	0.986	−0.04001	0.3256	0.972
Charlotte	−0.05186	0.4155	0.932	−0.05849	0.4483	0.960	−0.04520	0.3855	0.845
Chicago	−0.05291	0.3912	0.990	−0.05395	0.4043	0.996	−0.05524	0.4022	0.993
Cincinnati	−0.04124	0.4142	0.956	−0.05216	0.4560	0.940	−0.04559	0.4301	0.857
Cleveland	−0.04446	0.3321	0.955	−0.04742	0.3494	0.948	−0.04822	0.3474	0.967
Columbus	−0.05327	0.3899	0.982	−0.05376	0.3965	0.985	−0.05642	0.3966	0.950
Dallas-Fort Worth	−0.06492	0.4660	0.997	−0.05433	0.4296	0.994	−0.06434	0.4720	1.000
Denver	−0.02358	0.3811	0.958	−0.01959	0.3637	0.797	−0.02491	0.3952	0.975
Detroit	−0.04188	0.3472	0.963	−0.04312	0.3645	0.990	−0.04461	0.3578	0.967
Grand Rapids	−0.03937	0.3691	0.942	−0.03673	0.3678	0.970	−0.03019	0.3370	0.887
Hampton Roads	0.00290	0.2848	0.013	−0.04344	0.4506	0.784	−0.01853	0.3548	0.336
Hartford	−0.03171	0.3234	0.897	−0.03513	0.3474	0.936	−0.03771	0.3585	0.914
Honolulu	0.02964	0.2615	0.854	0.02782	0.2576	0.857	0.01495	0.3089	0.869
Houston	−0.05286	0.4491	0.997	−0.04889	0.4408	0.957	−0.06169	0.4805	0.992
Indianapolis	−0.05276	0.3845	0.972	−0.05768	0.4284	0.992	−0.05520	0.4062	0.983
Inland Empire	−0.02887	0.3887	0.989	−0.02046	0.3726	0.820	−0.01427	0.3546	0.892
Jacksonville	−0.02458	0.3748	0.827	−0.04517	0.4617	0.984	−0.04100	0.4283	0.985
Kansas City	−0.03909	0.3819	0.971	−0.04088	0.3921	0.999	−0.04352	0.3943	0.993
Las Vegas	0.00025	0.2842	0.001	0.00018	0.2899	0.025	−0.00528	0.2958	0.886
Los Angeles	−0.02052	0.3028	0.781	−0.03305	0.3471	0.926	−0.03425	0.3580	0.757
Louisville	−0.05151	0.4749	0.926	−0.03973	0.4275	0.931	−0.03255	0.4047	0.779
Memphis	−0.04381	0.4091	0.973	−0.04786	0.4432	0.988	−0.05315	0.4401	0.972
Milwaukee	−0.03570	0.3073	0.941	−0.04266	0.3424	0.954	−0.03521	0.2996	0.908

Table 5 (continued).

Metropolitan Statistical Area	1990			2000			2010		
	ζ	β	r^2	ζ	β	r^2	ζ	β	r^2
Minneapolis-St.Paul	−0.04722	0.3995	0.990	−0.05455	0.4281	0.987	−0.05764	0.4396	0.989
Nashville	−0.05033	0.4527	0.993	−0.05928	0.5242	0.996	−0.06460	0.5022	0.997
New Orleans	−0.03600	0.4076	0.767	−0.02686	0.3855	0.688	−0.01331	0.3600	0.204
New York	−0.03698	0.3636	0.981	−0.05176	0.4177	0.988	−0.04010	0.3564	0.942
Oklahoma City	−0.02432	0.3270	0.995	−0.04423	0.4079	0.998	−0.04293	0.3889	0.993
Orlando	−0.03329	0.4033	0.997	−0.03773	0.4017	0.988	−0.04051	0.4366	0.995
Philadelphia	−0.04931	0.3565	0.987	−0.05473	0.3922	0.978	−0.05044	0.3603	0.971
Phoenix	−0.03304	0.3675	0.997	−0.02684	0.3778	0.979	−0.03439	0.3815	0.974
Pittsburgh	−0.05723	0.4594	0.990	−0.06503	0.4929	0.985	−0.06105	0.4782	0.988
Portland	−0.03243	0.4762	0.856	−0.03178	0.4863	0.933	−0.03297	0.4986	0.964
Providence	−0.01163	0.2763	0.108	−0.02372	0.3179	0.476	−0.03402	0.3532	0.784
Raleigh	−0.05614	0.4095	0.984	−0.06599	0.4618	0.984	−0.06242	0.4393	0.962
Rochester, NY	−0.03945	0.3449	0.977	−0.03872	0.3366	0.986	−0.04054	0.3538	0.968
Sacramento	−0.01136	0.4047	0.909	−0.01090	0.4180	0.835	−0.00966	0.3742	0.678
Salt Lake City	0.00151	0.2976	0.052	0.00842	0.3261	0.677	0.00081	0.3234	0.011
San Antonio	−0.06932	0.5458	0.975	−0.06336	0.5216	0.990	−0.06376	0.5068	0.998
San Diego	0.00070	0.3423	0.017	−0.01039	0.3701	0.902	−0.02033	0.4143	0.947
San Francisco	−0.02030	0.4028	0.825	−0.03091	0.4398	0.872	−0.02632	0.4387	0.965
San Jose	−0.01012	0.3727	0.681	−0.00840	0.3810	0.471	−0.01027	0.3681	0.346
Seattle	−0.01773	0.4166	0.864	−0.00796	0.3589	0.860	−0.03205	0.4746	0.896
South Florida	−0.03926	0.3639	0.978	−0.03475	0.3543	0.878	−0.03886	0.3711	0.967
St. Louis	−0.04015	0.4398	0.930	−0.04502	0.4620	0.950	−0.05429	0.4965	0.995
Tampa Bay	−0.04075	0.3962	0.993	−0.04658	0.4173	0.997	−0.03513	0.3808	0.940
Washington	−0.05740	0.4936	0.992	−0.06451	0.5087	0.973	−0.05669	0.4755	0.952
Average	−0.03589	0.3860	0.854	−0.03913	0.4044	0.901	−0.03984	0.4021	0.882
σ_ζ	0.02019			0.02000			0.01878		

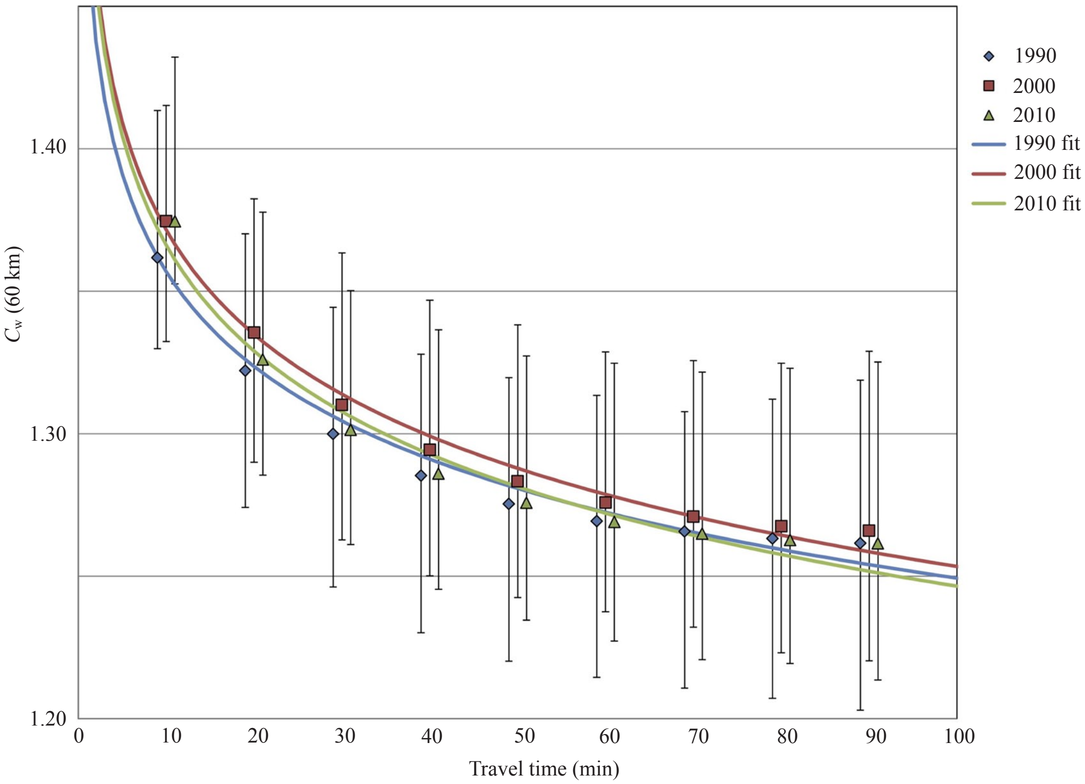


Figure 5. [In color online.] Circuity (C_w) by travel time, US average. An overlay of actual circuity data with plotted US averages showing the ability of equation (5) to model circuity.

Discussion

This paper quantifies road network circuity in MSAs across the United States. The results enable calculation of other measures, such as job accessibility (Levinson, 2012). The paper finds average weighted circuity increasing from 1.327 in 1990 to 1.339 in 2010. Given that 1.0 is the minimum possible circuity, this represents a 3.7% increase. Some areas such as Seattle, shaped by bodies of water, see values above 1.4. Over the period from 1990 to 2010 thirty-five of the United States' fifty-one most populated MSAs experienced statistically significant increases in circuity.

In the most populated MSAs in the United States, random points have not only become farther apart in distance, they are becoming more circuitous, suggesting that the more recently constructed parts of street networks are laid out less efficiently than older parts of the network. When this is weighted with actual trip distances in the US, the same holds true for many MSAs.

It is not surprising that new areas are less well connected than older areas, as that is part of the general process of network growth and infill (Levinson and Huang, 2011). But the trend must be that they stay less well connected, or that the amount of new network is becoming disproportionately significant.

Specific patterns of suburbanization (Baum-Snow, 2007; Jackson, 1987, Mieszkowski and Mills, 1993; Mubarak, 2004; Priemus et al, 2001; Susantono, 1998; Thurston and Yezer, 1994) are thus suspected causes of this circuity increase, and it is posited that this continued trend of progression toward areas with a more dendritic and hierarchical road network has caused the general increase in circuity over the past few decades. Another reason we might appear to observe this circuity increase is an improvement in the record keeping of suburban and rural roads, which are potentially less connected and may have been omitted in previous censuses.

While the study years identify an increasing trend, it is also possible that a peak home–work commute circuity occurred between 2000 and 2010.

Exploring the connections between intrametropolitan location and network structure and daily travel is an important to consider in subsequent studies. Determining causality, and whether influencing the circuity of an MSA could influence metropolitan economic productivity and agglomeration economies bears great potential for future research.

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